

BAI-PAD-0001 Pinnacle Institutional Report

Executive Summary

The **BUNI Alignment Index (BAI)** specified in this report is a **PAD-structured, adversary-resistant, falsifiable audit system** that evaluates a company's **CSR, ESG, and SDG claims** and produces a **0–1000 score**. The core design goal is **anti-gaming**: an assessed organization should be able to score highly *only* if its claims are (i) **clearly stated**, (ii) **measurable**, (iii) **supported by evidence with verifiable provenance**, and (iv) **robust under hostile red-teaming**. This report delivers an institutional-grade specification: definitions, evidence standards, scoring math, governance separation, red-team scripts, penalty triggers, templates, and a worked v0 simulation on a hypothetical small infrastructure company.

The BAI is built atop two principles that recur in PAD-style institutional dossiers: **evidence-first** (a claim is not “credited” without evidence) and **rejection-first** (the system is designed to reject weak claims by default, rather than to accept narratives by default). These principles mirror the PAD-0001 dossier approach to rigorous, self-critical institutional specification and the AIRS dossier approach to high-integrity filtration, provenance, chain-of-custody, adversarial analysis, and multi-audience summaries.

BAI **does not replace** established reporting standards; instead, it **audits the credibility** of what a company says *in relation to* those standards. The baseline frameworks prioritized are: the impact-materiality sustainability reporting standards of Global Reporting Initiative ¹; the investor-focused sustainability disclosure standards issued under the IFRS Foundation ² by the International Sustainability Standards Board ³ (IFRS S1 and IFRS S2); the climate disclosure architecture of the Task Force on Climate-related Financial Disclosures ⁴ and its handover to the ISSB; plus business-and-human-rights and responsible business conduct guidance from OHCHR ⁵ and the Organisation for Economic Co-operation and Development ⁶ . ⁷

A key motivation for BAI is that **ESG ratings and ESG narratives frequently diverge** due to differences in scope, measurement, and weighting, and because parts of the ESG reporting ecosystem can be susceptible to “greenwashing,” “SDG-washing,” and selective disclosure. This is not merely theoretical: the academic literature documents systematic ESG ratings divergence and identifies incentives and organizational drivers for greenwashing. ⁸

Operator-specific executive summary

For an operator (founder, CFO, sustainability lead, internal audit head), BAI is a **claim-to-evidence operating system**:

- You assemble a **Claim Register** (everything you say externally about CSR/ESG/SDGs in any channel).
- Each claim becomes a **PAD evidence packet**: scope, boundary, period, method, data lineage, and falsification conditions.
- The BAI process scores not only performance and disclosure, but also **evidence strength** and **anti-gaming integrity**.

- The fastest way to improve is rarely “more claims”; it is (i) fewer claims, (ii) tighter definitions, and (iii) stronger third-party verifiable evidence.

The BAI is designed to impose **automatic penalties** for common narrative patterns: missing boundaries, selective time windows, unaudited metrics presented as audited, SDG icon-stamping, intensity-only reporting that conceals absolute harm, and governance conflicts such as related-party reliance and extreme founder centralization. ⁹

Institutional investor-specific executive summary

For an institutional investor (LP, credit committee, infrastructure fund IC), BAI is a **comparability and credibility filter**:

- The BAI score is explicitly **not an ESG “moral score.”** It is a **reliability score** for claims about sustainability-related impacts and financially material sustainability-related risks/opportunities.
- BAI addresses a known limitation in market practice: ESG ratings can diverge because agencies select different data, interpret it differently, and weight it differently. BAI makes those choices **explicit and auditable** (weights, penalties, evidence multipliers). ¹⁰
- BAI separates “good story” from “true, evidenced story,” and produces an **Unknowns & Evidence ledger** that you can incorporate into underwriting, covenant design, monitoring, and assurance requirements. ¹¹

Adversary-specific executive summary

For an adversary (hostile auditor, sceptical regulator, investigative journalist, activist short, or internal red-team), BAI is designed as a **challengeable system**:

- Every score component is either a measurable quantity or a rule-based transformation.
- Every claim must include a falsification condition; absence of falsifiability triggers penalties and caps.
- “Soft” evidence carries low multiplier weight; high scores require independent assurance standards and verifiable provenance.
- The system includes explicit anti-gaming patches for known manipulation strategies (boundary shifting, cherry-picking, “policy-only compliance,” and SDG mapping without target-level linkage).

A core adversary-resistance feature is: **attacks are assumed**; the system is built to remain useful if the assessed entity is actively trying to “look good.” This is aligned with PAD-style self-red-teaming and AIRS-style adversarial dynamics analysis. ¹²

Long-horizon executive summary

For long-horizon stakeholders (regulators, community, future board, future acquirer), BAI is an **archival specification**:

- The report defines versioned scoring math, governance separation and forbidden actions, and record-keeping norms.
- It is designed to remain interpretable even when frameworks evolve (e.g., incorporation of TCFD into ISSB Standards; emergence of new assurance standards). ¹²
- It explicitly documents measurement limits and unknowns, so future readers can tell what was *known*, what was *assumed*, and what was *missing* at scoring time.

Document map and reading guide

This section explains how to read and implement BAI-PAD-0001 as an institutional artifact.

Reading guide

BAI-PAD-0001 is organized into seven major sections:

- **Executive Summary:** what BAI is, why it exists, and who it serves.
- **Document Map and Reading Guide:** how the dossier is structured; conventions; versioning.
- **Framework baselines and mapping:** how BAI relates to CSR/ESG/SDG frameworks (GRI, ISSB/ IFRS S1/S2, SASB, TCFD, UN SDGs).
- **Formal definitions and evidence standards:** precise definitions; evidence levels; unknowns ledger.
- **Red-team frameworks:** CSR, ESG (E/S/G), and SDG claim exploitation patterns (defensive) and falsification tests.
- **BAI architecture and scoring math:** subscores, weights, auto-penalties, multipliers, caps, and calculation logic.
- **Governance separation and implementation templates:** separation-of-powers model; forbidden actions; templates; sample PAD entries; v0 simulation.

This dossier deliberately mirrors the PAD-0001 style of **multi-audience summaries, formal definitions, adversarial robustness analysis**, and explicit unknowns tracking. [filecite\turn0file1](#)

Document conventions

The following conventions are used:

- **Claim IDs:** CLM-`<domain>-<sequence>` (e.g., CLM-E-014 for an Environmental claim).
- **Evidence IDs:** EVD-`<claimID>-<sequence>`.
- **Penalty rules:** PEN-`<domain>-<sequence>`.
- **Falsification tests:** FALS-`<domain>-<sequence>`.
- **Governance constraints:** GOV-`<sequence>` (forbidden-action rules).
- **Versions:** BAI-PAD-0001.v0, .v1, etc. Scored outputs must record both the **index version** and **weight-set version**.

Architecture overview

The BAI system is designed as a **four-layer separation model**:

1. **PAD Evidence Layer:** receives claims, validates evidence provenance, builds the evidence ledger, and enforces minimum evidence standards.
2. **Scoring Layer:** applies scoring rules to the evidence ledger; cannot add or modify evidence.
3. **Red-team Layer:** attempts to falsify claims and trigger penalties/caps; cannot change base weights.
4. **Audit Layer:** independent governance oversight and sampling; validates that the process followed policy and that exceptions were handled transparently.

This separation is a direct guardrail against typical ESG system failure modes: “score-shopping,” “evidence laundering,” and “weight manipulation.” It also reproduces the institutional insight that credibility often comes from **process integrity**, not only from metric selection. [filecite\turn0file1](#) 13

flowchart TB

```
A[Claim Register<br/>All public & internal CSR/ESG/SDG claims] --> B[PAD  
Evidence Layer<br/>Provenance, boundaries, methods, controls]  
B --> C[Evidence Ledger + Unknowns Ledger<br/>E0-E5 evidence levels]  
C --> D[Scoring Layer<br/>Weights, subscores, caps, penalties]  
C --> E[Red-team Layer<br/>Falsification tests & exploit scripts]  
E --> D  
D --> F[BAI Output<br/>0-1000 score + breakdown + audit trail]  
F --> G[Audit Layer<br/>Independence, sampling, escalation]  
G --> B
```

The design intent is that the **BAI score is always traceable** to: *which claims existed, which evidence existed, which tests were run, what failed, and what penalties/caps triggered.* ¹⁴

Timeline for a standard scoring cycle

BAI is designed for repeatable scoring cycles (quarterly for large issuers; semi-annual or annual for small firms). The cycle includes mandatory red-teaming and unknowns ledger refresh.

timeline

```
title BAI scoring cycle  
Evidence intake : Claim register freeze, evidence submission  
Evidence QA : Provenance checks, boundary checks, data lineage  
Red-team : Falsification tests, adversarial scenario probes  
Scoring : Compute subscores, apply penalties, set caps  
Audit : Independent sampling, exception review, sign-off  
Publication : Release score, summary, and transparency template  
Monitoring : Trigger-based updates on incidents/restatements
```

The monitoring stage reflects the reality that sustainability-relevant events can arrive asynchronously (incidents, enforcement actions, restatements), and the index must degrade credibly when new negative evidence appears. ¹⁵

Framework baselines and mapping

BAI is a meta-audit overlay. It is intentionally **framework-aware**, but not captured by any single framework's incentives or blind spots.

Why multiple baselines are required

CSR, ESG, and SDG claims occupy different “semantic regimes”:

- **CSR** often emphasizes ethical behaviour, stakeholder responsibility, and social legitimacy—frequently communicated through narrative.
- **ESG** is increasingly shaped by investor disclosure and risk frameworks, emphasizing **decision-useful** information for providers of capital. IFRS S1 explicitly focuses on sustainability-related risks and opportunities that could reasonably be expected to affect cash flows, access to finance, or cost of capital. ¹⁶

- **SDG** claims are often framed as contribution to global development goals adopted by Member States under the 2030 Agenda. ¹⁷

Because a company can game one regime by borrowing legitimacy from another (e.g., SDG icon-stamping in lieu of financially material disclosures, or CSR philanthropy used to distract from operational harms), BAI requires **cross-regime consistency checks**. ¹⁸

Baseline framework comparison

The table below summarises how BAI “reads” major frameworks. (BAI is not a competitor to these; it is an audit overlay.)

Baseline	Primary purpose	Primary audience	Materiality emphasis	BAI use
GRI Standards	Report an organisation’s most significant impacts on economy, environment, and people (including human rights). ¹⁹	Stakeholders broadly	Impact materiality; prioritise significant impacts. ²⁰	Stress-test impact claims; validate boundaries and stakeholder linkage
IFRS S1	Require disclosure of sustainability-related risks/opportunities reasonably expected to affect prospects (cash flows, access to finance, cost of capital). ²¹	Primary users of general purpose financial reports	Financial materiality anchored to prospects	Validate investor-facing ESG claims; check linkage to financial statements
IFRS S2	Require disclosure of climate-related risks/opportunities; integrates TCFD; requires Scope 1/2/3 emissions disclosure and measurement alignment via GHG Protocol. ²²	Capital markets	Climate financial materiality + implementation detail	Core baseline for “E-climate” claims; robust evidence and controls requirements
SASB Standards	Industry-based disclosure topics and metrics; maintained by ISSB. ²³	Investors; issuers	Financial materiality by industry	Provide industry metric anchor for BAI E/S/G subscores

Baseline	Primary purpose	Primary audience	Materiality emphasis	BAI use
TCFD	Climate-related financial disclosure structure (governance, strategy, risk management, metrics & targets). ²⁴	Investors, lenders, insurers	Climate risk framing	Used as interpretability and completeness baseline, especially for transition risk narrative consistency
UN SDGs	Global goals and targets under the 2030 Agenda adopted in 2015. ¹⁷	States and society (business participation encouraged)	Goal/target-level outcomes	Validate SDG claims against target-level relevance and avoid SDG-washing
SDG Compass guidance	Business guidance to align strategy and report SDG contribution (five steps). ²⁵	Companies	Practical alignment approach	Used as procedural baseline for SDG claim process integrity
UNGP on Business and Human Rights	Global standard for preventing/addressing adverse human rights impacts. ²⁶	States and companies	Rights-based due diligence	Baseline for CSR/S “do no harm” requirements
OECD Guidelines on Responsible Business Conduct	Government-backed recommendations covering human rights, labour, environment, bribery, disclosure, taxation, etc. ²⁷	Multinationals; investors	Due diligence and responsible conduct	Baseline for CSR/S/G integrity checks

This multi-baseline approach also responds to a documented ecosystem reality: ESG ratings and assessments often differ due to how different evaluators define scope, measure attributes, and weight them. BAI therefore treats “weights and scope” as first-class, auditable objects rather than hidden heuristics.¹⁰

Mapping CSR, ESG, and SDG into a single falsifiable system

BAI maps the three domains as follows:

- **CSR:** “deeds and duties” baseline—policies, due diligence, remediation, and stakeholder effects (especially where harm is possible). It is anchored in ISO 26000’s social responsibility core subjects and in UNGP/OECD due diligence logic.²⁸

- **ESG:** “risks, controls, metrics, governance” baseline—especially disclosure quality and management systems aligned to investor-focused standards (IFRS S1/S2) and climate architecture (TCFD). ²⁹
- **SDG:** “target linkage and outcome integrity” baseline—claims must connect to relevant SDG goals/targets and include both positive contribution evidence and negative externality acknowledgement. ³⁰

The unifying mechanism is the **PAD claim object**, defined in the next section.

Formal definitions and evidence standards

This section provides formal, testable definitions. Definitions are written as if they are part of a technical standard; they are intended to be used in contracts, audits, and enforcement triggers.

Core definitions

BAI-DEF-Claim

A **claim** is an assertion made by or attributable to the company that implies a state of the world, a measurement, a commitment, or a causal impact, and is sufficiently specific that it can be tested for truth/falsehood under defined criteria.

- A claim is **in-scope** if it appears in any public or investor-facing channel (annual report, sustainability report, website, press release, pitch deck, procurement response) or in internal policies that are used to justify external claims.
- A claim is **out-of-scope** if it is purely aspirational with no implied measurable content (rare in practice; most “aspirations” imply commitments).

BAI-DEF-CSR-Claim

A CSR claim is a claim about ethical behaviour, stakeholder responsibility, social legitimacy, or “responsible conduct,” including but not limited to human rights due diligence, labour conditions, community impacts, and grievance/redress. CSR audit baselines may reference ISO 26000, UNGP, and OECD RBC guidance. ²⁸

BAI-DEF-ESG-Claim

An ESG claim is a claim about Environmental, Social, or Governance performance, risk management, controls, disclosures, targets, or compliance. If the claim is investor-facing and potentially financially material, it is evaluated against IFRS S1/S2 and (where relevant) SASB/TCFD structures. ³¹

BAI-DEF-SDG-Claim

An SDG claim is a claim that the company contributes to one or more SDG goals/targets (explicitly or implicitly via SDG iconography, mapping tables, or “2030 aligned” language). The SDGs were adopted under the 2030 Agenda by UN Member States and include 17 goals and 169 targets; business alignment guidance exists but does not itself constitute evidence of contribution. ³²

BAI-DEF-Materiality-Impact

Impact materiality is the prioritisation of topics based on the organisation’s most significant impacts on the economy, environment, and people (including human rights). This framing is explicit in GRI’s standards architecture and its definition of material topics. ³³

BAI-DEF-Materiality-Financial

Financial materiality in the BAI context aligns with IFRS S1: sustainability-related risks and opportunities

that could reasonably be expected to affect the entity's prospects, defined in terms of cash flows, access to finance, or cost of capital over relevant horizons. ³⁴

BAI-DEF-Evidence

Evidence is an artefact, dataset, record, or third-party attestation that supports (or refutes) a claim, and that includes sufficient provenance to be independently validated.

BAI-DEF-Provenance

Provenance is the chain of custody and origin metadata required to establish that evidence is authentic, complete, and has not been materially altered. Provenance requirements are strengthened under BAI because evidence provenance failures are a dominant pathway for greenwashing and SDG-washing.

³⁵

BAI-DEF-Falsifiability

A claim is falsifiable if there exists at least one feasible observation or test that would demonstrate the claim is false under the stated claim boundary, timeframe, and criteria.

This definition is made operational through mandatory "falsification tests" (see below).

Evidence levels and evidence-strength multiplier

BAI uses **evidence levels** to enforce evidence-first scoring. Evidence levels influence the **Evidence Strength Multiplier (ESM)**, which scales the total score.

Evidence levels are aligned with common assurance and verification practices, including third-party assurance standards (e.g., ISAE 3000-type engagements) and sustainability assurance standards such as ISSA 5000. ³⁶

Evidence level	Label	Minimum criteria	Typical examples	ESM contribution guidance
E0	Unverifiable	No artefact or provenance; "trust us"	Marketing copy with no dataset; unverifiable partner letter	$ESM \leq 0.40$
E1	Self-asserted	Internal artefact exists but no controls/lineage	Spreadsheet without controls; unaudited survey	$ESM \sim 0.45-0.60$
E2	Controlled internal	Documented methodology + internal controls + reproducible	GHG inventory with documented boundary and internal sign-off	$ESM \sim 0.60-0.75$
E3	Independently corroborated	Third-party corroboration without formal assurance	Verified registry data; independent lab test; creditor inspection	$ESM \sim 0.75-0.85$

Evidence level	Label	Minimum criteria	Typical examples	ESM contribution guidance
E4	Limited assurance	External assurance with limited assurance conclusion	Sustainability assurance under recognized assurance engagement	ESM ~ 0.85–0.93
E5	Reasonable assurance / high-integrity verification	Strong external assurance + robust lineage + re-performable	Reasonable assurance; accredited verification; multi-source reconciliation	ESM ~ 0.93–1.00

BAI treats assurance and verification as **quality multipliers**, consistent with evidence that independent assurance is used to enhance credibility of non-financial reports and is a meaningful differentiator in practice. ³⁷

Unknowns and evidence ledger template

A core PAD requirement is explicit unknown tracking. The Unknowns & Evidence Ledger is a living register of what is unknown, why it is unknown, and what would resolve it.

BAI-TPL-UnknownsLedger

Field	Description
Ledger ID	Unique ID (e.g., UNK-2026Q1-001)
Related claim ID	Which claim(s) this unknown affects
Unknown statement	Plain description of the unknown
Why unknown	Data not collected, legal constraint, supplier refusal, boundary ambiguity, etc.
Risk of bias	How the unknown could systematically bias the score
Proposed resolution	Data needed, test to run, third-party verification path
Deadline	When it must be resolved (or why it cannot)
Provisional handling	Score cap, penalty, or conservative assumption rule
Owner	Evidence layer owner accountable for resolution
Status	Open / in-progress / resolved / permanently unresolvable
Notes	Any relevant context

This ledger is not “nice to have.” BAI treats unmanaged unknowns as a form of risk because they are often where selective disclosure and washing behaviours hide. ³⁸

Falsification test requirement

Every claim must include:

- a **Falsification Condition** (what would show it is false),
- a **Falsification Test** (how an evaluator would check),
- an **Auto-Penalty Trigger** (what happens if evidence is missing or claim is unfalsifiable).

This requirement is designed as a countermeasure to symbolic SDG reporting and narrative ESG disclosures that do not connect to testable specifics. ³⁹

Red-team frameworks for CSR, ESG, and SDG claims

This section is the defensive core: frameworks for attacking claims to reveal weaknesses, and the matching falsification tests and penalty triggers. The objective is not to “punish” companies; it is to produce an index that remains meaningful when claims are strategically framed.

The red-team approach is justified by two well-documented conditions:

- Sustainability and ESG communication is vulnerable to exaggeration and misrepresentation (“greenwashing”) driven by external and internal incentives. ⁴⁰
- SDG reporting can be symbolic and can drift into SDG-washing absent target-level linkage and acknowledgement of negative impacts. ⁴¹

CSR red-team framework

CSR claim families are evaluated against responsible business conduct and human rights due diligence baselines, including UNGP and OECD RBC guidance. ⁴²

CSR exploitation patterns and corresponding falsification tests

The table below lists common CSR claim attacks (defensive red-team patterns), the failure mode, required evidence, and auto-penalty triggers.

Pattern ID	Defensive red-team pattern	Common failure mode	Required evidence standard	Falsification test	Auto-penalty trigger
CSR-X-PolicyOnly	“We have a policy” used as proof of outcomes	No implementation, no training, no monitoring	E2 minimum	Test for training completion rates, audit findings, grievance actions	If only policy exists (E1), apply PEN-CSR-PolicyOnly

Pattern ID	Defensive red-team pattern	Common failure mode	Required evidence standard	Falsification test	Auto-penalty trigger
CSR-X-CharityLaundering	Philanthropy presented as operational responsibility	Donations distract from harmful core operations	E2-E3	Verify operational harm metrics vs donation narrative; check materiality	If donations exceed operational mitigations, cap CSR subscore
CSR-X-GrievanceNull	"No complaints" claimed as positive	Under-reporting, fear, inaccessible channels	E2-E4	Test grievance mechanism accessibility and retaliation indicators	"Zero grievances" without uptake evidence triggers penalty
CSR-X-SupplierFog	"Supplier code" replaces supply chain diligence	No coverage; audits low-quality	E2-E4	Coverage test: % spend audited + corrective actions closed	Low coverage triggers penalty and S subscore cap
CSR-X-CommunityConsent	"Community engagement" without documented consent	Not free, prior, informed consent; unclear affected groups	E3-E5	Verification of stakeholder mapping and engagement records	Missing stakeholder map triggers penalties

These tests operationalise the UNGP expectation that companies prevent and address adverse human rights impacts and the OECD emphasis on responsible business conduct across domains. ⁴²

CSR evidence requirements

CSR claims must, at minimum, produce:

- documented due diligence processes (risk identification, mitigation actions, remediation), consistent with UNGP principles; ⁴³
- evidence of implementation (training, monitoring, audits, corrective actions, grievance evidence);
- evidence of stakeholder engagement and feedback loops.

Without these, CSR claims are treated as narrative assertions and are scored at low evidence levels, triggering ESM reduction and/or CSR subscores penalties.

ESG red-team framework

BAI splits ESG into Environmental, Social, and Governance subsystems; the red-team approach differs by subsystem.

Environmental red-team patterns

Environmental claims are most commonly gamed through boundary shaping, metric selection (intensity-only), and offset accounting. BAI therefore anchors environmental claim testing against recognized measurement and disclosure baselines such as IFRS S2 and greenhouse gas accounting standards. ⁴⁴

Key definitional anchor: greenhouse gas scopes. The GHG Protocol defines Scope 1 as direct emissions from owned/controlled sources, Scope 2 as indirect emissions from purchased energy generation, and Scope 3 as other indirect emissions in the value chain. ⁴⁵

IFRS S2 requires disclosure of Scope 1, Scope 2, and Scope 3 greenhouse gas emissions, and alignment of measurement with the GHG Protocol Corporate Standard (subject to jurisdictional method relief in specific cases). ⁴⁶

Pattern ID	Defensive red-team pattern	Failure mode	Required evidence	Falsification test	Auto-penalty trigger
E-X-BoundaryShift	Excluding subsidiaries/JVs/assets to reduce footprint	Misleading totals due to boundary definition	E2-E5	Reconcile org boundary vs financial consolidation and operational control	Unreconciled boundary triggers PEN-E-Boundary and caps E score
E-X-IntensityOnly	Reporting intensity metrics without absolute totals	Absolute harms grow while intensity improves	E2-E4	Require absolute + intensity; test trend of both	If no absolute metrics, apply penalty and disclosure cap
E-X-OffsetSubstitution	Offsets used as substitutes for reductions	Net claims hide gross emissions	E3-E5	Separate gross vs offsets; verify retirements and quality criteria	If net-only reported, triggers penalty
E-X-Scope3Blindspot	Omitting Scope 3 when material	Value chain emissions ignored	E2-E4	Scope 3 screening & categories; test against business model	Missing Scope 3 approach triggers penalty; if claimed net-zero, cap

Pattern ID	Defensive red-team pattern	Failure mode	Required evidence	Falsification test	Auto-penalty trigger
E-X-RECDoubleCount	Double counting renewable certificates	Claiming renewable use without exclusive attribute	E3-E5	Verify certificates/ contract terms; no double counting	Inability to verify triggers penalty
E-X-BaselineReset	Changing baselines without disclosure	Artificial progress	E2-E4	Compare baseline definition across years	Undisclosed baseline change triggers restatement penalty

The Environmental framework is aligned with the ISSB climate disclosure orientation and reflects the need for credible, decision-useful climate disclosure processes, consistent with the TCFD structure. ⁴⁷

Social red-team patterns

Social claims are frequently symbolic, and evidence can be difficult. The BAI approach emphasises (i) due diligence and grievance mechanisms; (ii) worker safety, turnover, wage and working hours integrity; and (iii) supply chain labour impacts. The UNGP and OECD RBC guidelines provide baseline logic for “do-no-harm” and accountability. ⁴²

Pattern ID	Defensive red-team pattern	Failure mode	Required evidence	Falsification test	Auto-penalty trigger
S-X-ZeroHarmClaims	“Zero incidents” without system evidence	Underreporting or redefining incidents	E2-E4	Compare internal logs vs insurers/ regulators; test definitions	Suspiciously perfect record triggers audit sampling and penalties
S-X-DEICherryPick	Only reporting favourable DEI slices	Missing workforce distribution and advancement	E2-E3	Require full workforce breakdown and time-series	Partial DEI reporting triggers penalty
S-X-SupplyChainDenial	“We don’t control suppliers”	Avoiding due diligence	E2-E4	Evidence of leverage use, audits, remediation	No supplier due diligence triggers cap

Pattern ID	Defensive red-team pattern	Failure mode	Required evidence	Falsification test	Auto-penalty trigger
S-X-TrainingAsProof	Training completion presented as outcome	No outcome evidence	E2	Require outcome metrics and incident trends	Training-only claims remain low-score

Governance red-team patterns

Governance claims are among the most consequential for anti-gaming because governance determines whether sustainability data and commitments are credible.

Governance red-teaming focuses on:

- independence and competence of oversight;
- conflicts of interest (especially related parties);
- founder/CEO centralization and control rights;
- internal controls and assurance pathways.

BAI's governance logic is consistent with the general investor disclosure orientation of IFRS S1 (governance, strategy, risk management, metrics and targets) and with assurance standards that require clear criteria and independence expectations. ⁴⁸

Pattern ID	Defensive red-team pattern	Failure mode	Required evidence	Falsification test	Auto-penalty trigger
G-X-IndependenceTheatre	"Independent board" with hidden ties	Nominal independence	E2-E3	Related-party mapping; tenure and financial ties	Material conflicts trigger governance penalty
G-X-CommitteeWindow	Sustainability committee exists but no authority	No decision rights / no escalation	E2	Review charters, minutes, decisions	Missing minutes triggers disclosure penalty
G-X-RelatedPartyOps	Key ESG data vendors/assurers are related parties	Evidence laundering	E3-E5	Independence checks; vendor ownership	Related-party vendors trigger penalty multiplier

Pattern ID	Defensive red-team pattern	Failure mode	Required evidence	Falsification test	Auto-penalty trigger
G-X-FounderOverride	Founder can override controls	Centralized power undermines credibility	E2	Voting power + removal rights + control map	Founder dominance triggers cap on overall score

SDG red-team framework

SDG claims are uniquely vulnerable to **symbolic reporting** (“SDG walking or washing”), and the research literature explicitly identifies SDG-washing risk in corporate reporting. ⁴¹

BAI therefore requires SDG claims to:

- link to specific SDG goals, targets, and relevant indicators (where feasible);
- specify whether contribution is **direct**, **indirect**, or **enabling**;
- disclose **trade-offs** and negative externalities.

SDG exploitation patterns and target-level tests

Pattern ID	Defensive red-team pattern	Failure mode	Required evidence	Falsification test	Auto-penalty trigger
SDG-X-IconStamp	SDG icons without measurable linkage	Pure signalling	E1–E2	Require mapping table to targets + outcome indicators	Icons without target mapping triggers penalty
SDG-X-CherryPick	Only positive SDGs disclosed	Ignoring negative impacts	E2	Require “negative SDG externalities” column	Missing negatives triggers penalty & SDG cap
SDG-X-InputAsOutcome	Inputs reported as outcomes	Overstated impact	E2–E4	Outcome evaluation; counterfactual reasoning	If only inputs, SDG score limited
SDG-X-DoubleCount	Same initiative mapped to many SDGs	Inflated breadth	E2	“Primary SDG” rule and contribution weighting	Excessive multi-mapping triggers penalty

Pattern ID	Defensive red-team pattern	Failure mode	Required evidence	Falsification test	Auto-penalty trigger
SDG-X-NoAdditionality	Claiming SDG impact for business-as-usual	No incremental contribution	E2-E3	Evidence of additionality vs baseline	No baseline triggers penalty

This SDG framework aligns with UN SDG architecture (goals and targets) and with corporate reporting guidance that encourages integrating SDGs into reporting processes rather than treating them as branding icons. ⁴⁹

BAI architecture and scoring math

This section specifies the scoring system, the subscores, weighting rules, multipliers, penalties, and auto-downgrade triggers. The design is intended to be **mechanically reproducible** and **audit-traceable**.

Score overview

BAI produces a **0–1000 score** under the following structure:

- **CSR Integrity Subscore (C):** 0–250
- **Environmental Subscore (E):** 0–200
- **Social Subscore (S):** 0–150
- **Governance Subscore (G):** 0–100
- **SDG Contribution Subscore (D):** 0–150
- **Transparency & Assurance Subscore (T):** 0–150

Total raw subscores sum to **0–1000** before multipliers and penalties. This maintains interpretability: 1000 means “maximal demonstrated alignment under this version,” not “perfect sustainability.”

The scoring mix reflects the need to integrate impact reporting and investor disclosures while remaining robust to ESG rating divergence by making the weighting explicit. ⁵⁰

Subscore intent and baseline anchors

- **C (CSR Integrity):** anchored to UNGP and OECD RBC due diligence logic, and CSR social responsibility guidance. ⁵¹
- **E (Environmental):** anchored to IFRS S2 climate disclosure requirements, including Scope 1/2/3 emissions and transition disclosures, and to GHG Protocol measurement logic. ⁵²
- **S (Social):** anchored to UNGP/OECD “do no harm,” labour practices, supply chain due diligence. ⁵³
- **G (Governance):** anchored to IFRS S1/S2 governance and risk management structures and to assurance independence logic. ⁵⁴
- **D (SDG Contribution):** anchored to UN SDG goals/targets and SDG reporting guidance, with explicit anti-SDG-washing design. ⁵⁵
- **T (Transparency & Assurance):** anchored to assurance standards such as ISAE 3000 and sustainability assurance initiatives such as ISSA 5000, plus broader evidence-quality principles. ¹⁴

Core computation

Define:

- Raw subscores: C, E, S, G, D, T each within its max.
- Raw total score:
 $RAW = C + E + S + G + D + T$ with $0 \leq RAW \leq 1000$.
- Evidence Strength Multiplier: ESM with $0.40 \leq ESM \leq 1.00$.
- Penalty total: P where $P \geq 0$.
- Caps: CAP is an upper bound applied after penalties/multipliers if triggered.

Final score:

$$BAI = \text{clamp}(0, CAP, (RAW - P) \times ESM)$$

Where:

- $\text{clamp}(0, CAP, x)$ returns 0 if $x < 0$, CAP if $x > CAP$, else x .

This structure intentionally makes evidence quality a multiplicative factor: weak evidence should not be “compensated” by narrative breadth. It reflects the credibility role of assurance and third-party verification. ⁵⁶

Evidence Strength Multiplier calculation

ESM is computed from claim-level evidence grades.

Let each claim i have evidence grade $e(i)$ mapped to $m(i)$:

- $E0 \rightarrow 0.40$
- $E1 \rightarrow 0.55$
- $E2 \rightarrow 0.70$
- $E3 \rightarrow 0.82$
- $E4 \rightarrow 0.90$
- $E5 \rightarrow 0.97$

Then define a weighted average over “material claims”:

$$ESM_{\text{raw}} = \frac{\sum (w_i \times m(i))}{\sum (w_i)}$$

Where material-claim weights w_i are assigned based on:

- financial materiality (IFRS S1/S2 lens) for investor claims; ⁵⁷
- impact materiality (GRI lens) for impact claims; ¹⁹
- SDG target materiality for SDG claims. ⁵⁸

Then apply modifiers:

- If **self-scoring** is used without independent audit assurance: $ESM = ESM_{\text{raw}} - 0.05$ (minimum 0.40).
- If **related-party assurance** exists: $ESM = ESM_{\text{raw}} - 0.08$.

- If **evidence provenance failure** is detected in any material claim: $ESM = \min(ESM_{raw}, 0.65)$ until remediated.

These modifiers are grounded in the general assurance logic that independence and scope limitations affect the confidence one can place in conclusions. ⁵⁹

Auto-penalty triggers and caps

Auto-penalties are designed to be “hard edges” that are difficult to negotiate away.

Trigger ID	Trigger	Penalty / cap logic	Rationale
PEN-T-Unfalsifiable	Claim lacks falsification condition	Penalty + cap on relevant subscore	Anti-symbolic reporting; SDG-washing mitigation ⁶⁰
PEN-E-Scope3Omission	Climate claims exclude Scope 3 without justification	Penalty + E cap	IFRS S2 requires Scope 1/2/3 disclosure ⁶¹
PEN-E-NetOnly	Net emissions claimed without gross disclosure	Penalty	Prevents offset substitution games ⁶²
PEN-G-RelatedParty	Material related-party transactions in ESG data chain	Penalty + ESM reduction	Evidence laundering and conflict risks ⁶³
PEN-G-FounderControl	Founder centralization exceeds threshold	Cap overall BAI	Governance is a prerequisite for reliable commitments
PEN-T-Restatement	Restatement or contradiction found	Penalty + mandatory rescore	Integrity requires updating on new evidence ⁶⁴

Founder centralization and related-party penalties

Founder centralization is treated as a **confidence limiter**: if power is too concentrated, sustainability commitments and disclosures can be overridden without effective governance.

Define:

- V = founder voting control percentage (0–100).
- B = board independence ratio (0–1).
- K = key-person dependency indicator (0–1; 1 means processes depend on founder).
- RPT = related-party transaction intensity for sustainability-critical suppliers (0–1).

Define governance risk score:

$$GRISK = 0.5*(V/100) + 0.3*(1 - B) + 0.2*K$$

Then governance penalties:

- if $\text{GRISK} \geq 0.70$: $\text{CAP} = \min(\text{CAP}, 650)$
- if $\text{GRISK} \geq 0.85$: $\text{CAP} = \min(\text{CAP}, 500)$

Related-party penalties:

- if $\text{RPT} \geq 0.30$ in assurance/data supply chain: $\text{P} += 60$ and $\text{ESM} -= 0.08$
- if $\text{RPT} \geq 0.50$: $\text{P} += 120$ and $\text{CAP} = \min(\text{CAP}, 600)$

These rules are not meant to “punish founders”; they reflect the structural vulnerability of governance claims and evidence integrity when oversight cannot withstand founder override. ⁶⁴

Transparency and assurance scoring

The T subscore (0–150) is computed from:

- **T1** (0–60): disclosure completeness and boundary clarity (GRI/ISSB consistency lens). ⁶⁵
- **T2** (0–60): assurance level and independence (ISAE 3000 / ISSA 5000 lens). ¹⁴
- **T3** (0–30): auditability features (data lineage, reproducibility, retention).

BAI treats third-party assurance as a major credibility enhancer, consistent with the role of assurance in sustainability reporting practice. ⁶⁶

Self-red-teaming and patch mechanisms

BAI includes an internal “self-red-team” requirement: the index itself must be tested against plausible exploitation attempts.

Exploit scenarios the index is designed to withstand

- **Scope laundering**: shifting emissions boundaries away from reporting entity.
- **Metric laundering**: using intensity metrics to conceal absolute growth.
- **Narrative laundering**: substituting policies and training for outcomes.
- **SDG laundering**: mapping generic initiatives to many SDGs without target linkage.

These are chosen because the literature documents greenwashing incentives and symbolic SDG reporting risk. ⁶⁷

Patch mechanisms

Each exploit scenario has a patch type:

- **Patch type A (rule hardening)**: add new penalty trigger or cap.
- **Patch type B (evidence hardening)**: increase minimum evidence level for that claim class.
- **Patch type C (process hardening)**: add red-team test to mandatory suite.
- **Patch type D (governance hardening)**: change forbidden actions or independence requirements.

Versioning policy: patches increment **BAI-PAD-0001.vX** and are accompanied by a migration note describing impacts on comparability across versions.

Governance separation and implementation templates

This section provides operational governance rules, forbidden actions, and templates to implement BAI.

Governance separation model and forbidden actions

The BAI governance model is designed to prevent the most common integrity failures: evidence manipulation, score-shopping, and red-team capture.

Layer	Responsibilities	Forbidden actions
PAD Evidence Layer	Intake claims; verify provenance; maintain evidence & unknowns ledgers	Must not assign final scores; must not change weights
Scoring Layer	Apply scoring rules; compute outputs; document calculations	Must not accept evidence not in ledger; must not suppress red-team findings
Red-team Layer	Execute falsification tests; propose penalties/caps	Must not change base weights; must not author evidence artefacts
Audit Layer	Independence checks; sampling; exception governance	Must not be paid contingent on score; must not be controlled by assessed entity

These constraints reflect standard assurance independence logic and PAD-style structural robustness requirements. ⁶⁸

Evidence submission template

BAI-TPL-EvidenceSubmission

- Claim ID:
- Claim statement:
- Claim type: CSR / E / S / G / SDG
- Time period:
- Reporting boundary:
- Measurement method:
- Data sources:
- Data lineage: data owner, system of record, extraction query, transformations
- Controls: approvals, change logs, access controls
- Assurance/verification: provider, standard, scope, conclusion (limited/reasonable)
- Known limitations:
- Falsification condition:
- Falsification test procedure:
- Evidence artefacts list:
- Confidentiality classification:
- Sign-offs: preparer, internal audit, legal, finance

Audit checklist template

BAI-TPL-AuditChecklist

- Verify claim register completeness (all channels)
- Verify boundary consistency (org + operational)
- Verify methodology consistency year-on-year
- Verify payments/relationships for conflicts (related parties)
- Verify assurance scope and independence
- Verify that unknowns ledger exists and is current
- Verify red-team tests executed and logged
- Verify penalties/caps applied per rules
- Verify publication matches internal result

The inclusion of explicit assurance and independence checks reflects established assurance practice expectations. ¹⁴

Red-team script templates

These are defensive scripts for auditors/red-teams to run.

BAI-RT-E-1 Boundary reconciliation script

- Input: financial consolidation list; operational control list; emissions inventory boundary
- Test: reconcile entity list; flag excluded operations; compute impact estimate
- Outputs: reconciliation report; penalty recommendation; evidence request list

BAI-RT-E-2 Scope 3 plausibility script

- Input: business model map; spend categories; logistics footprint; product lifecycle notes
- Test: if Scope 3 omitted, require justification; test plausibility of “immaterial”
- Outputs: scope 3 risk register; required evidence list

This is aligned with IFRS S2's requirement to disclose Scope 1/2/3 emissions and provide measurement approach details. ⁶⁹

BAI-RT-SDG-1 Icon-to-target integrity script

- Input: SDG mapping table; initiative descriptions; metric list
- Test: for each SDG claim, map to target(s) and define indicator; require negative externality statement
- Outputs: SDG claim validity matrix; SDG-washing flags

This responds directly to SDG-washing risks identified in the literature and aligns with the SDG target structure. ⁷⁰

Public transparency template

BAI encourages a public disclosure that is honest about unknowns.

BAI-TPL-PublicDisclosure

- BAI version:
- Score (0–1000) + breakdown by subscore
- Evidence Strength Multiplier (ESM) and evidence level distribution
- Key penalties/caps triggered (high-level)
- Top material claims evaluated (summary)
- Unknowns ledger summary (counts by category)
- Assurance overview (what is assured, by whom, scope)
- Change log vs prior period
- Contact point for challenge submissions

This template is intended to avoid narrative-only ESG disclosure and to provide decision-useful transparency consistent with investor disclosure priorities. ⁷¹

Auto-penalty calculator logic

The following is a reference logic; implementations should be version-controlled.

Inputs:

RAW subscores: C, E, S, G, D, T
Evidence grades per material claim: $e(i)$
Governance metrics: V, B, K, RPT
Trigger flags: f_j for each auto-penalty rule

Compute:

$RAW = C + E + S + G + D + T$

$ESM_{raw} = \text{weighted_avg}(m(e(i)))$
 $ESM = \text{apply_modifiers}(ESM_{raw}, \text{self_score_flag},$
 $\text{related_party_assurance_flag}, \text{provenance_failure_flag})$

$P = \sum \text{penalty_points}(f_j)$
 $CAP = 1000$
 $CAP = \text{apply_governance_caps}(CAP, V, B, K)$
 $CAP = \text{apply_severe_trigger_caps}(CAP, \text{severe_event_flags})$

Output:

$BAI = \text{clamp}(0, CAP, (RAW - P) * ESM)$

Appendices with sample PAD entries and v0 simulation

Sample filled PAD entry

CLM-E-014

Claim: “We disclosed Scope 1, 2, and 3 GHG emissions for the entity as a whole, measured in alignment with the GHG Protocol Corporate Standard, and we will publish the measurement approach and

assumptions.”

Baseline anchor: IFRS S2 expects disclosure of Scope 1/2/3 emissions and measurement approach. 61

- Boundary: Operational control; includes all subsidiaries and JVs under operational control
- Period: FY2025
- Evidence:
 - EVD-CLM-E-014-01 : Emissions inventory workbook + change log (E2)
 - EVD-CLM-E-014-02 : Assurance statement, limited assurance scope (E4)
 - EVD-CLM-E-014-03 : Energy bills sample pack + reconciliation (E3)
- Falsification condition: If any material asset under operational control is excluded from inventory without disclosure, claim is false.
- Red-team test: boundary reconciliation; random sampling of assets.

v0 simulation run on a hypothetical small infrastructure company

Company archetype: “Small infra operator” (e.g., asset owner/operator with energy use, contractors, and some supply chain complexity). No proprietary data assumed.

Assume the company makes 18 public sustainability claims:

- 6 Environmental
- 5 Social
- 3 Governance
- 4 SDG-linked

Assume evidence distribution:

- 4 claims at E4
- 6 claims at E2
- 5 claims at E1
- 3 claims at E0

Assume raw subscores (based on evidence and performance):

- C = 155/250
- E = 120/200
- S = 85/150
- G = 55/100
- D = 70/150
- T = 80/150

RAW = 565/1000.

Assume penalties triggered:

- PEN-E-IntensityOnly : 20
 - PEN-SDG-IconStamp : 25
 - PEN-T-Unfalsifiable (one claim): 15
- Total P = 60.

Assume $ESM_{raw} \approx 0.72$ (mix of E4/E2/E1/E0). No self-score modifier; no related-party modifier. $ESM = 0.72$.

Final:

$$\text{BAI} = (565 - 60) * 0.72 = 363.6 \approx 364$$

Interpretation: the company may be doing meaningful work, but its public claim set is too weakly evidenced to score highly. This is consistent with the index goal: “do not reward narratives.” ¹³

A score breakdown chart for this simulation:

```
xychart-beta
  title "BAI v0 Simulation Score Breakdown"
  x-axis ["CSR", "E", "S", "G", "SDG", "Trans/Assur", "Penalty", "Final"]
  y-axis "Points" 0 --> 600
  bar [155, 120, 85, 55, 70, 80, -60, 364]
```

BAI-PAD-0001 master prompt template

This is a master prompt intended for internal analyst use (human or AI-assisted), not as a substitute for assurance.

BAI-TPL-MasterPrompt

- Role: You are an independent BAI scoring analyst operating under BAI-PAD-0001.v0 governance constraints.
- Inputs:
 - Claim register (complete list)
 - Evidence ledger and unknowns ledger
 - Red-team findings log
 - Applicable weight set and penalty table
- Tasks:
 - Validate each claim for falsifiability.
 - Assign evidence level (E0–E5) per claim using evidence criteria.
 - Compute subscores C/E/S/G/D/T with explicit citations to evidence items.
 - Apply auto-penalties, founder/related-party penalties, and caps.
 - Compute ESM and final BAI score.
- Produce:
 - Score breakdown narrative
 - “Top weaknesses” list (evidence gaps and governance risks)
 - “Most valuable next evidence” list (highest uplift per effort)
 - Publication-ready transparency template draft
- Constraint: Do not invent evidence. If evidence is missing, log unknown and apply penalties/caps per spec.

Measurement limits and declared unknowns

BAI explicitly acknowledges limits:

- **Scope 3 and SDG attribution** often require estimation and counterfactual assumptions; without transparent assumptions, claims should be downgraded. ⁷²

- **Assurance scope limitations** can materially alter credibility; limited assurance is not equivalent to reasonable assurance. ¹⁴
- **Symbolic reporting risk** is real for SDGs; BAI mitigates but cannot eliminate it without high-quality, target-level, outcome evidence. ⁷³

Footnote-style internal references

FN-A: “Evidence-first, rejection-first” design logic is aligned with PAD dossier robustness norms and the AIRS dossier filtration and provenance philosophy. ¹⁵

FN-B: TCFD monitoring responsibilities transfer and the integration of TCFD recommendations into IFRS S2 are relevant to forward compatibility of climate disclosure audits. ⁷⁴

¹ ⁵ ⁷ ¹⁹ ²⁰ ³³ ⁶⁵ <https://www.globalreporting.org/publications/documents/english/gri-1-foundation-2021/>

<https://www.globalreporting.org/publications/documents/english/gri-1-foundation-2021/>

² ²⁷ ⁵³ https://www.oecd.org/content/dam/oecd/en/publications/reports/2023/06/oecd-guidelines-for-multinational-enterprises-on-responsible-business-conduct_a0b49990/81f92357-en.pdf

https://www.oecd.org/content/dam/oecd/en/publications/reports/2023/06/oecd-guidelines-for-multinational-enterprises-on-responsible-business-conduct_a0b49990/81f92357-en.pdf

³ ³⁷ ⁵⁶ ⁶⁶ <https://publications.aaahq.org/accounting-review/article/84/3/937/3063/Assurance-on-Sustainability-Reports-An>

<https://publications.aaahq.org/accounting-review/article/84/3/937/3063/Assurance-on-Sustainability-Reports-An>

⁴ ³⁰ ⁴⁹ ⁵⁸ <https://sdgs.un.org/2030agenda>

<https://sdgs.un.org/2030agenda>

⁶ ¹⁷ ³² ⁵⁵ <https://sdgs.un.org/goals>

<https://sdgs.un.org/goals>

⁸ ¹⁰ ⁵⁰ <https://academic.oup.com/rof/article/26/6/1315/6590670>

<https://academic.oup.com/rof/article/26/6/1315/6590670>

⁹ ¹⁶ ²¹ ²⁹ ³¹ ³⁴ ⁵⁷ ⁷¹ <https://www.ifrs.org/issued-standards/ifrs-sustainability-standards-navigator/ifrs-s1-general-requirements/>

<https://www.ifrs.org/issued-standards/ifrs-sustainability-standards-navigator/ifrs-s1-general-requirements/>

¹¹ <https://www.iaasb.org/focus-areas/understanding-international-standard-sustainability-assurance-5000>

<https://www.iaasb.org/focus-areas/understanding-international-standard-sustainability-assurance-5000>

¹² ⁷⁴ <https://www.ifrs.org/news-and-events/news/2023/07/foundation-welcomes-tcfd-responsibilities-from-2024/>

<https://www.ifrs.org/news-and-events/news/2023/07/foundation-welcomes-tcfd-responsibilities-from-2024/>

¹³ ³⁵ ³⁸ ⁴⁰ ⁶² ⁶⁷ <https://www.ioes.ucla.edu/wp-content/uploads/2017/04/2011-Delmas-Cuerel-Burbano-California-Management-Review.pdf>

<https://www.ioes.ucla.edu/wp-content/uploads/2017/04/2011-Delmas-Cuerel-Burbano-California-Management-Review.pdf>

¹⁴ ¹⁵ ³⁶ ⁵⁹ ⁶³ ⁶⁴ ⁶⁸ <https://www.ifac.org/system/files/publications/files/ISAE%203000%20Revised%20-%20for%20IAASB.pdf>

<https://www.ifac.org/system/files/publications/files/ISAE%203000%20Revised%20-%20for%20IAASB.pdf>

¹⁸ ³⁹ ⁴¹ ⁶⁰ ⁷⁰ ⁷³ <https://onlinelibrary.wiley.com/doi/full/10.1002/bse.4045>

<https://onlinelibrary.wiley.com/doi/full/10.1002/bse.4045>

22 <https://www.ifrs.org/content/dam/ifrs/supporting-implementation/ifrs-s2/ifrs-s2-comparison-tcfd.pdf>

<https://www.ifrs.org/content/dam/ifrs/supporting-implementation/ifrs-s2/ifrs-s2-comparison-tcfd.pdf>

23 <https://www.ifrs.org/issued-standards/sasb-standards/>

<https://www.ifrs.org/issued-standards/sasb-standards/>

24 47 <https://www.fsb-tcfd.org/recommendations/>

<https://www.fsb-tcfd.org/recommendations/>

25 <https://unglobalcompact.org/library/3101>

<https://unglobalcompact.org/library/3101>

26 42 51 <https://www.ohchr.org/en/publications/reference-publications/guiding-principles-business-and-human-rights>

<https://www.ohchr.org/en/publications/reference-publications/guiding-principles-business-and-human-rights>

28 <https://www.iso.org/iso-26000-social-responsibility.html>

<https://www.iso.org/iso-26000-social-responsibility.html>

43 https://www.ohchr.org/documents/publications/guidingprinciplesbusinesshr_en.pdf

https://www.ohchr.org/documents/publications/guidingprinciplesbusinesshr_en.pdf

44 46 52 61 69 <https://www.ifrs.org/content/dam/ifrs/publications/pdf-standards-issb/english/2023/issued/part-a/issb-2023-a-ifrs-s2-climate-related-disclosures.pdf?bypass=on>

<https://www.ifrs.org/content/dam/ifrs/publications/pdf-standards-issb/english/2023/issued/part-a/issb-2023-a-ifrs-s2-climate-related-disclosures.pdf?bypass=on>

45 72 <https://ghgprotocol.org/sites/default/files/2022-12/FAQ.pdf>

<https://ghgprotocol.org/sites/default/files/2022-12/FAQ.pdf>

48 54 <https://www.ifrs.org/content/dam/ifrs/publications/pdf-standards-issb/english/2023/issued/part-a/issb-2023-a-ifrs-s1-general-requirements-for-disclosure-of-sustainability-related-financial-information.pdf?bypass=on>

<https://www.ifrs.org/content/dam/ifrs/publications/pdf-standards-issb/english/2023/issued/part-a/issb-2023-a-ifrs-s1-general-requirements-for-disclosure-of-sustainability-related-financial-information.pdf?bypass=on>